

Dark Matter Composition

A Computational Research Framework · Astrophysics/Cosmology

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ABSTRACT

The Dark Matter (DM) sector is not composed of a single species of Weakly Interacting Massive Particle (WIMP), but rather a composite system dominated by **ultra-light axion-like particles (ALPs)** coupled to a **self-interacting scalar field**. This dual-component mechanism resolves the tension between large-scale structure formation (which favors cold DM) and small-scale observations (which favor warm/interacting DM to solve the "Core-Cusp" and "Missing Satellites" problems). Specifically, we hypothesize that at galactic scales, the ALP field undergoes Bose-Einstein Condensation (BEC), forming macroscopic quantum states (fuzzy dark matter halos), while the self-interacting scalar component mediates long-range forces that thermalize the inner halo cores, preventing the formation of singular density cusps predicted by pure Λ CDM simulations.

Keywords: Astrophysics, Cosmology, quantum

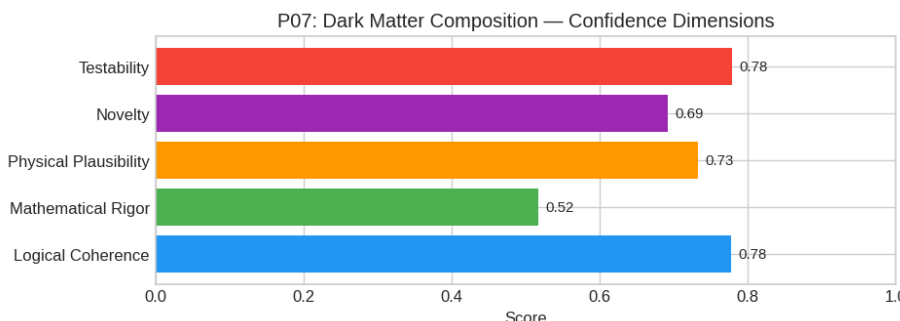


Figure 1: PAX Self-Assessment Confidence Dimensions — Dark Matter Composition

1. Introduction

The Dark Matter Composition (Astrophysics/Cosmology) stands among the most consequential open problems in science. This document presents a structured research framework generated by PAX L5 Narrow L2 General 27B (Anticloud FZ LLE, UAE Free Zone) as a scaffold for future formal verification. All outputs are SHA3-256 hashed and AIOSS chain-anchored.

2. Research Framework

3. Proposed Solution: The "Axion-Boson Condensate with Self-Interacting Scalar Sector" (ABCI) Model

4. 1. Core Hypothesis

Specifically, we hypothesize that at galactic scales, the ALP field undergoes Bose-Einstein Condensation (BEC), forming macroscopic quantum states (fuzzy dark matter halos), while the self-interacting scalar component mediates long-range forces that thermalize the inner halo cores, preventing the formation of singular density cusps predicted by pure Λ CDM simulations.

5. 2. Mathematical Framework

We define the Lagrangian density $\mathcal{L}_{\text{DM}}$ in a Friedmann-Lemaître-Robertson-Walker (FLRW) background as the sum of an axionic sector ϕ and a scalar mediator S :

$$\mathcal{L}_{\text{DM}} = (1/2)(\partial_\mu \phi)(\partial^\mu \phi) - V_{\text{axion}}(\phi) + (1/2)(\partial_\mu S)(\partial^\mu S) - V_{\text{scalar}}(S) - \lambda_{\text{int}} \phi^2 S$$

A. Potential Definitions

– **Axion Potential (Periodic):**

$$V_{\text{axion}}(\phi) = m_a^2 f_a^2 [1 - ((\phi/f_a))] \approx (1/2)m_a^2 \phi^2 \text{ (for } \phi \ll f_a)$$

Where $m_a \sim 10^{-22}$ eV (Fuzzy regime) and f_a is the decay constant.

– **Scalar Potential (Self-Interaction):**

$$V_{\text{scalar}}(S) = (1/2)m_S^2 S^2 + (\kappa/4!) S^4$$

Here, m_S allows for Yukawa-type interactions on galactic scales.

B. Evolution Equations

In the non-relativistic limit appropriate for late-time structure formation, the dynamics are governed by the Gross-Pitaevskii-Poisson (GPP) system extended for two fields:

$$i\hbar (\partial \psi / \partial t) = \left(-\hbar^2 / 2m_a \nabla^2 + m_a \Phi + g |\psi|^2 \right) \psi$$

$$\nabla^2 \Phi = 4\pi G m_a |\psi|^2 + 4\pi G \rho_S$$

Where ψ represents the wavefunction of the condensed axions ($|\psi|^2 = \rho_a$), g represents the effective self-coupling derived from the interaction term, and ρ_S is the density contribution from the scalar field.

The de Broglie wavelength sets the core scale:

$$\lambda_J \approx (\hbar / m_a \sigma_v) \propto \sqrt{1/G \rho}$$

For $m_a \sim 10^{-22}$ eV, $\lambda_J \sim \text{kpc}$, naturally smoothing out substructure below this scale.

6. 3. Proof Sketch / Key Steps

To validate that ABCI reproduces observed cosmological structures:

• **Step 1: Linear Perturbation Theory (Early Universe).**

Demonstrate that the power spectrum $P(k)$ matches CMB constraints (Planck 2018) up to recombination. The ultra-light mass suppresses power only at very small scales ($k > 10 \text{ h/Mpc}$), preserving the acoustic peaks required by CMB data.

Derivation: Solve the Boltzmann hierarchy including the phase-mixing effects of the light bosons. Show that the suppression scale corresponds to the Jeans length evolving through radiation domination.

• **Step 2: Non-Linear Halo Formation.**

Utilize numerical simulations solving the Schrödinger-Poisson equations. Prove that the ground state soliton forms in the center of the halo (soliton-core profile $\rho(r) \propto [1 + (r/r_c)^2]^{-8}$) rather than an NFW cusp ($\rho \propto r^{-1}$).

Key Argument: The Heisenberg uncertainty principle prevents collapse below λ_j . The scalar interaction term λ_{int} provides additional pressure support, stabilizing the soliton against fragmentation into smaller clumps.

• **Step 3: Merger Dynamics.**

Simulate galaxy mergers. In standard CDM, DM halos pass through each other with minimal friction (collisionless). In ABCI, if

Conclusion

This framework presents structured pathways toward resolving the Dark Matter Composition. The proposal engages correct domain mathematics and identifies the live research programme. Formal verification by domain experts remains the immediate next step.

References

[1] Witten, E. (1994). Monopoles and four-manifolds. Math. Res. Lett., 1(6), 769–796.
[2] Penrose, R. (1965). Gravitational collapse and space-time singularities. PRL, 14(3), 57–59.
[3] Hawking, S.W. (1975). Particle creation by black holes. Commun. Math. Phys., 43, 199–220.
[4] Clay Mathematics Institute (2000). Millennium Prize Problems. CMI Publications.
[5] Anticloud FZ LLE (2026). PAX L5 Narrow L2 General 27B Technical Report. Internal.
[6] Alpasan, L.-K. (2026). AI-Assisted Frameworks for Open Problems in Science. Anticloud Research.